



Molecular characterization and in-depth genomic analysis to unravel the pathogenic features of an environmental isolate *Enterobacter* sp. S-33

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Abstract

Enterobacter species represent widely distributed opportunistic pathogens, commonly associated with plants and humans. In the present study, we performed a detailed molecular characterization as well as genomic study of a type VI secretion system (T6SS) bacterium belonging to member of the family *Enterobacteriaceae* and named *Enterobacter* sp. S-33. The comparative sequence analysis of the 16S rRNA gene showed that the strain was closely related to other *Enterobacter* species. The complete genome of the strain with a genome size of 4.6 Mbp and GC-content of 55.63% was obtained through high-quality sequencing. The genomic analysis with online tools unravelled the various genes belonging to the bacterial secretion system, antibiotic resistance, virulence, efflux pumps, etc. The isolate showed the motility behavior that contributes to *Enterobacter* persistence in a stressed environment and further supports infections. PCR amplification and further sequencing confirmed the presence of drug-efflux genes *acrA*, *acrB*, and outer membrane genes, viz. *OmpA*, *OmpC*, and *OmpF*. The cell surface hydrophobicity and co-aggregation assay against different bacterial strains illustrated its putative pathogenic nature. Genome mining identified various biosynthetic gene clusters (BGCs) corresponding to non-ribosomal proteins (NRPS), siderophore, and arylpolyene production. Briefly, genome sequencing and detailed characterization of environmental *Enterobacter* isolate will assist in understanding the epidemiology of *Enterobacter* species, and the further prevention and treatment of infectious diseases caused by these broad-host range species.

Keywords T6SS · Genome · AMR · Virulence · OMPs

Introduction

Enterobacter genus was first reported by Hormaeche and Edwardsi in 1960, and until now, 58 species and 7 subspecies are included in this genus. Members of the *Enterobacter* genus are opportunistic pathogens belonging to the ESKAPE category and are responsible for infection in the lower respiratory tract and bacteremia (Davin-Regli et al. 2019). These are considered potent pathogens for the healthcare and human gastrointestinal tract, the most common

reservoir for these bacteria (Ren et al. 2010). The identified strains from various geographical regions as well as from patients showed resistance against a wide range of antibiotics including aminoglycosides, tetracyclines, fluoroquinolones, β -lactams, trimethoprim, and finally become multidrug-resistant (MDR) strains. For example, *Enterobacter cloacae* strain R11, isolated from sewage water, showed tolerance to antibiotic polymyxin (up to 192 μ g/ml) which is higher than any reported for other Gram-ve bacteria (Zhong et al. 2017). The genome analysis of *E. cloacae* EC_38V1M1 identified various genes associated with resistance to multiple antibiotics including the metallo- β -lactamase (Villa et al. 2013). It is clearly evident that genome analysis will reveal the genetic features that directly or indirectly affect its corresponding biological activity and other important information that will help to identify specific genes involved in plant growth-promoting mechanisms, endocrine systems, surface attachment, and other related metabolic mechanisms (Asaf et al. 2020).

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Enterobacter bacteria have also been reported as plant pathogens (Abedinzadeh et al. 2023) and plant growth promoters in agriculture (Mezzatesta et al. 2012). Recently, it was demonstrated that these species pose a threat to plants by causing internal decay of onion, yellowing disease of papaya, and lucerne seeds (García-González et al. 2018). *Enterobacter* sp. SA187, an endophytic proteobacterium isolated from the desert plant *Indigofera argentea*, enhanced *Arabidopsis thaliana* growth under abiotic stressors. The genome analysis identified the genes involved in chemotaxis, plant colonization, oxidative stress, production of antimicrobial compounds, etc. (Andrés-Barrao et al. 2016). Molecular analyses have shown that the *Enterobacter cloacae* complex (ECC) contains 6 species; *E. cloacae* subsp. *cloacae*, *E. asburiae*, *E. kobei*, *E. hormaechei*, *E. ludwigii*, and *E. nimipressuralis*, and possesses various potential virulence factors, such as secretion systems and multiple drug resistance efflux systems (Iredell et al. 2016; Paauw et al. 2008). The evolution of whole genome sequencing technologies has been providing major improvements in the epidemiological characterization of pathogens of great public health importance by developing typing methods for bacterial phylogeny, tracking and investigation of outbreaks, and monitoring of antimicrobial resistance (Gilmour et al. 2013).

To survive in their natural habitat, bacteria compete with many other microorganisms for available resources and space. Many Gram-negative bacteria use a specialized protein secretion system known as the “type VI secretion system” (T6SS) to eliminate their target co-habiting bacteria in a contact-dependent manner (Chassaing and Cascales 2018; Coulthurst 2019). The T6SS is versatile machinery used by bacterial cells to target host cells or rival bacteria, and is structurally and functionally similar to a bacteriophage tail (Douzi et al. 2018; Basler et al. 2012; Silverman et al. 2012). T6SS has 13 (TssA to TssM) conserved core components which are predicted to form contractile machinery encoding a baseplate, VgrG spike, VipA/B contractile outer sheath, and Hcp inner tube in both pathogenic and non-pathogenic bacteria (Cascales and Cambillau 2012; Silverman et al. 2012). The functional roles of T6SS in virulence, immunomodulation, persistence, adherence, invasion, and inter-bacterial competition have been demonstrated (Gallique et al. 2017; Repizo et al. 2015; Russell et al. 2014). The finding of the secretion system in bacteria is important for the study of bacterial pathogenesis and virulence activity (Wang et al. 2019). These secreting cells have specific immunity proteins that protect the secreting cells from antibacterial effectors (Russell et al. 2013).

Bacterial cell surface hydrophobicity represents the adhesion ability of the bacteria and is considered an important determinant of pathogenicity. It is measured by evaluating the bacterial affinity for hydrocarbon solvents and bacteria with higher hydrophobicity bind firmly to epithelial cells

and thereby influence adhesion ability (Busscher and van der Mei 2012). Bacterial adhesion augments the pathogenicity (Busscher and van der Mei 2012) and is escorted by surface protein receptors, surface exopolysaccharides, S-layer protein, mucin binding protein (MucBP), lipoteichoic acid, etc. The adhesion processes are affected by various environmental factors like pH, media, and ionic concentration which influence the bacterial surface charge (Habimana et al. 2014). Similarly, auto-aggregation and co-aggregation are characteristics that provide certain advantages to the microorganism in colonizing the gastro-intestinal tract and also represent the pathogenic behavior of an organism (Pan et al. 2017).

Some isolates of *Enterobacter* spp. showed tolerance to several antibiotics which reflects the accumulation of resistance mechanisms (Boutarfi et al. 2019). The outer membrane of Gram-ve bacteria facilitates the passive movement of hydrophilic drugs (< 600 kDa) and allows the selective influx of solutes into the cell (Pagès et al. 2008). The proteins forming these pores are referred to as outer membrane proteins (OMPs) or porins (Fernández and Hancock 2012), and are found in many different forms such as β -barrel membrane anchors (OmpA, OmpX), non-specific porins (OmpC, OmpF), and substrate-specific porins (PhoE, LamB). These OMPs impart many physiological roles in bacteria (Lin et al. 2002).

Besides antibiotic resistance, OMPs enhance bacterial attachment to mucosal surfaces, invasion, and resistance to antimicrobial peptides, thereby playing a key role in bacterial pathogenesis (Confer and Ayalew 2013). OmpX neutralizes the host defense mechanism, whereas OmpC and OmpF are responsible for the influx of antibiotics and other solutes. The deletion mutants of OmpA and OmpX reduced the adhesion and invasion to human epithelial cells in *Cronobacter sakazakii* (Kim et al. 2010). Similarly, the OmpC and OmpF mutations reduced its invasive, adhesive, and colonization abilities (Hejair et al. 2017). These pieces of evidence suggest that OMPs coordinate multifunctional roles including antibiotic resistance, virulence properties, and promote adaptation to adverse conditions (Sato et al. 2017). Many of the efflux genes such as *acrA*, *acrB*, etc., have been characterized in *Enterobacter* sp., *E. cloacae*, and *E. ludwigii* (Fernández and Hancock 2012).

With the expansion of our knowledge about threats posed by various *Enterobacter* pathogens, it is necessary to elucidate the genomic features and gene repository of newly identified strains. In the present study, a bacterial strain belonging to *Enterobacteriaceae* was isolated from the soil sample. The presence of T6SS genes, virulence factors, and efflux pumps was analyzed by polymerase chain reaction and further confirmed by sequencing. To unravel the presence of virulence gene repertoires and antimicrobial resistance genes (AMRs) in an environmental isolate, genome

sequencing was performed. Additionally, genome analysis will be helpful in understanding the different metabolic pathways that potentially contribute to survival, stress tolerance, and the biotechnological application of an isolated strain.

Material and methodology

Sample collection and microorganism

The soil sample was collected from the BIT Mesra campus, Ranchi, Jharkhand, India (23.41° N, 85.43° E). It was brought to the laboratory in ziplock bags and stored in cool conditions until processing. The soil sample was serially diluted with sterile 0.9% NaCl (w/v) solutions and dilutions were spread on Luria-Bertani (LB)-agar (Himedia, India). All plates were incubated at 37 °C under static condition for overnight. Morphologically distinct colonies were randomly picked and streaked on the LB-agar plate. The isolates were preserved as glycerol stock (20% v/v) at – 80 °C. All the colonies were further streaked on McConkey agar for the isolation of lactose-fermenting *Enterobacter* spp. Among 20 streaked colonies, one colony showed a change in color, and it was further analyzed for detailed characterization.

Biochemical characterization

The test isolate was characterized for various biochemical features. Gram staining was performed by a Gram staining kit (Himedia, India). Test for starch hydrolysis, IMViC (indole, methyl red, Voges Proskauer, citrate utilization), cellulase, pectinase, lipase, and catalase assays were performed following standard protocol (Harley and Prescott 2002). The KB009-HiCarbo Kit was used for the carbohydrate fermentation test as per manufacturer protocol. The antagonistic activity against the bacterial pathogens such as *Bacillus subtilis*, *Salmonella typhi*, *Escherichia coli*, and *Staphylococcus aureus* and fungal strains *Aspergillus niger*, *Microsporum gypseum*, *H. gypsium*, and *Penicillium citrium* was determined by the well diffusion method (Singh et al. 2015).

Antibiotics sensitivity test

The disk diffusion method was used against selected isolate to check their susceptibility to different antibiotics as recommended by the Clinical and Laboratory Standards Institute (CLSI 2012). The isolate was grown in LB broth overnight in the shaker, after that the selected isolate was spread in an agar plate and then the paper disk containing the antibiotic was placed in the agar plate for diffusion and incubated for 24–48 h. The results were observed by measuring the diameter of the zone of inhibition created by the antibiotic disk

against the tested isolates. A list of antibiotics was used as erythromycin (15 µg), ampicillin (10 µg), kanamycin (30 µg), tetracycline (30 µg), ciprofloxacin (5 µg), gentamycin (10 µg), fluconazole (25 µg), streptomycin (10 µg), vancomycin (30 µg), and voriconazole (10 µg). These are the most common antibiotics used against both environmental and clinical strains.

Motility test

The culture was freshly grown on an LB-agar plate for motility test. For the swimming test, culture was spot inoculated on solidified tryptone swim plate (1% tryptone, 0.5% NaCl, 0.3% agar) and incubated for 16 h at 25 °C. For swarming, spot inoculation was performed on dextrose-containing swarming media (0.5% bacto-agar, 8 g L⁻¹ nutrient broth, 5 g L⁻¹ dextrose) and incubated for 24 h at 30 °C. For twitching, LB agar (1% agar) media was used for stab inoculation and inoculated plates were incubated at 30 °C for 24–48 h. After incubation, the presence of a turbid circular zone indicated swimming, movement of inoculation for swarming, and circular turbid zone for twitching activity (Singh et al. 2015).

Abiotic stress tolerance test

The growth of selected isolate S-33 was examined for its ability to tolerate abiotic stress conditions such as pH (5 to 10), temperature (25 °C to 50 °C), and NaCl (sodium chloride) (5 to 10%) in LB-broth by measuring OD at 600 nm in a spectrophotometer (Biophotometer, Eppendorf, USA) and uninoculated medium was used as a blank. For the pH tolerance study, the pH of LB-medium was adjusted between 5 and 10 with a suitable pH buffer. A 100 µL of fresh bacterial culture was transferred in 5 mL of LB broth medium comprising different pH and kept for shaking at 180 rpm at 37 °C. The growth of the culture was measured after 24 h at 600 nm. To test the temperature tolerance, a 100 µL of bacterial suspension was transferred in 5 mL of LB broth and tubes were incubated at different temperatures between 30 and 45 °C for 24 h at 180 rpm. Similarly, for NaCl tolerance, 5 mL of LB-broth medium supplemented with 5, 6, 7, 8, 9, and 10% NaCl were inoculated with 100 µL of bacterial suspension, and kept for 24 h at 180 rpm for recording the growth.

Molecular identification and phylogenetic analysis

The 16S rRNA gene was amplified by polymerase chain reaction (PCR) using a pair of universal primers, 27F1 and 1494 Rc (Table 1), as previously described (Li et al. 1993). The reaction was carried out in a thermal cycler (Applied Biosystems, USA), run in 1% agarose gel prepared in 1× Tris-acetate buffer (TAE) and further visualized in a

Table 1 List of primers used in this study

Sl. no.	Primer name	DNA sequence (5'→3')
1	27F1	AGAGTTTGATCCTGGCTCAG
	1492R	GGTTACCTTGTTACGACTT
2	Hcp-F	TCCCGACCGATAACAACACC
	Hcp-R	GATGTCGTGCATCAGGGGAT
3	VgrG-F	TGAGCGTGTTGTGCGAAA
	VgrG-R	TGACGCCCCGTAATATCCTGC
4	acrA-F	TGATGCTCTCAGGCAGCTTA
	acrA-R	GCCTGGATATCGCTACCTTC
5	acrB-F	CGTCTCCATCAGCGACATTAAC
	acrB-R	GAACCGTATTCCTCAACGCGA
6	OmpA-F	GGATGATAACGAGGCGCAAA
	OmpA-R	CAACCAGATGTCTACGCTGAAG
7	OmpC-F	AGGGTTAATCAGTAAGCAGTGG
	OmpC-R	AGGGTGAATTGTAAGAACCGAA
8	OmpF-F	GATCTGTACGGGAAAGCAGTT
	OmpF-R	GCCGAAGCCCTGTTTCATTA

Gel-documentation system (Biorad, USA). The 1.5 kb PCR product was purified by PCR purification kit (Qiagen, Germany) and sent for sequencing at Genotypic Labs Pvt. Ltd. (Genotypic, India). The obtained sequence was submitted to GenBank and accession number was assigned. The phylogenetic relationship was established by using the 16S rRNA sequence of S-33 with the type strains of the *Enterobacter* genus.

T6SS gene characterization

The genomic DNA of selected isolate S-33 was extracted by a bacterial genomic DNA extraction kit (Qiagen, Germany) according to manufacturer instructions. For the T6SS gene identification, the marker genes Hcp (hemolysin cytotregulated protein) and VgrG (valine-glycine repeat protein G) gene were amplified by the suitable primer sets (Table 1). The condition involved initial denaturation at 94 °C for 3 min, followed by 30 cycles of denaturation at 94 °C for 30 s, annealing at 54 °C for 30 s, extension for 1 min at 72 °C, and a final extension at 72 °C for 10 min in the thermal cycler (Applied Biosystems, USA). The partially amplified product was purified using a QIAquick PCR purification kit (QIAGEN, Germany) and sequenced at Eurofins Genomics Pvt. Ltd. (Eurofins, India). The sequence was submitted to the Genbank database and phylogenetic relationship was established.

Screening of OMP genes by multiplex PCR

To detect the presence of OMP genes, viz. OmpA, OmpC, and OmpF, hexaplex PCR was performed using suitable

oligonucleotides (Table 1). The reaction was performed in a 50 µL reaction mixture that contained 10 µL of (5×) Gotaq Flexi buffer (Promega, USA), 1 µL of dNTP mixture (Promega, USA), 1.25 µL of forward and reverse primer (10 mM), 1.25 µL of template DNA, and 0.1 µL (100 U) of Gotaq flexi DNA polymerase (Promega, USA) in a thermal cycler (Applied Biosystems, USA). The reaction cycle was performed with an initial denaturation at 94 °C for 2 min, followed by 35 cycles of denaturation at 94 °C for 45 s, annealing temperature of 53 °C for 45 s, and an extension at 72 °C for 45 s, with a final extension at 72 °C for 10 min. The PCR product was gel extracted, purified, and sequenced at Eurofins Genomics Pvt. Ltd. (Eurofins, India).

Detection of efflux genes

The detection of efflux genes (acrA and acrB) in strain S-33 was performed by multiplex PCR using suitable primer sets (Table 1). The reaction mixture contained 1.0 µL template DNA, 0.5 µL (10 pmol) of forward and reverse primer sets, 2.5 mM MgCl₂, 1.5 µL (1.5 U) of Taq DNA polymerase, 1 µL dNTPs (200 µM), 2 µL 10× PCR buffer (75 mM) Tris-HCl, pH 9.0, 2 mM MgCl₂, 50 mM KCl, 20 mM (NH₄)₂SO₄, and sterile distilled water up to 25 µL. The PCR condition for gene amplification was performed as follows: initial denaturation at 94 °C for 5 min, followed by 30 cycles with denaturation at 94 °C for 1 min, annealing at 58 °C for 1 min, extension at 72 °C for 1 min, and final extension at 72 °C for 10 min.

Evaluation of hydrophobicity and co-aggregation assay

To determine the cell surface hydrophobicity, *Enterobacter* sp. S-33 was grown up to OD 0.5 at 600 nm. The obtained cells were washed twice with PBS (phosphate-buffered saline) buffer (pH 7.0) and 3 mL of the washed cells were mixed with 1 mL of solvents like hexane, toluene, xylene, and hexadecane. The suspension was vortexed for 5 min and incubated at room temperature for 30 min under static condition. The adherence of S-33 cells to the solvents was qualitatively assessed following the difference in OD₆₀₀ before and after the assay (Collado et al. 2008). For the co-aggregation assay, cell suspensions of S-33 and *B. subtilis*, *S. typhi*, *E. coli*, and *S. aureus* with an OD₆₀₀ 1.0 was prepared, and mixed in an equal ratio in the tubes, and vortexed thoroughly. The tubes were incubated at 37 °C in static mode and absorbance was recorded at 2 h, 4 h, 6 h, and 24 h. The percentage of co-aggregation was determined following the formula used by Collado et al. (2008).

Whole genome sequencing

The whole genome sequencing was performed by a hybrid assembly approach for assembling the bacterial genome using the Unicycler tool at Genotypic Technology Pvt. Ltd (Genotypic, India). Library construction was performed using the QIASeq FX DNA library preparation protocol. The 50 ng of DNA was enzymatically fragmented, end-repaired, and A-tailed in a one-tube reaction using the FX Enzyme Mixture provided with the QIASeqDX DNA Kit. The end-repaired and adenylated fragments were subjected to adapter ligation to generate a sequencing library. The library was subjected to 6 cycles of indexing-PCR with an initial denaturation at 98 °C for 2 min, cycling at 98 °C for 20 s, 60 °C for 30 s, 72 °C for 30 s, and final extension at 72 °C for 1 min to enrich the adaptor-tagged fragments. The amplified library was purified and quantified following standard protocol. The raw data was demultiplexed using Bcl2fastq software and FastQ files were generated based on the unique dual barcode sequence (Andrews 2010). The initial contig level assembly is generated from Illumina short reads, followed by the use of Nanopore long reads to build longer scaffolds by finding the overlap between generated contigs, which often allows it to resolve all repeats in the genome, resulting in a complete genome assembly. (<https://github.com/rrwick/Unicycler#options-and-usage>). The RAST (Rapid Annotation using Subsystem Technology) server was used to annotate the open reading frames and BLAT v2.0 to validate the predictions.

ANI (average nucleotide identity) analysis

To explore the genetic distance and relatedness, ANI was performed for the genome sets containing S-33 and type strains of *Enterobacter* genomes using ANI-BLAST (ANiB) by using the software JSpecies v.1.2.1 (<http://imedea.uib-csic.es/jspecies/>) with a threshold of 95% as the cut-off for species (Kumari et al. 2023). The occurrence of an ANiB value equal to or more than 95% represents the strains belonging to the same species (Lalucat et al. 2020). Additionally, in silico DNA–DNA hybridization (isDDH) was used to assess the genetic relationships at the genomic level.

Gene ontology and pathway analysis

Gene prediction of S-33 was performed using the draft genome in the Prokka v1.14 tool (Seemann 2014) and gene ontology annotation was predicted against the Uniprot Bacterial Database using Diamond Blast Program (Buchfink et al. 2015). The pathway analysis was carried out using the KAAS database (Moriya et al. 2007). The protein sequences were used as input for metabolic pathway analysis and the database provides functional annotation of genes by

BLAST comparisons against the manually curated KEGG (Kyoto Encyclopedia of Genes and Genomes database). The genome sequence was analyzed on the antiSMASH server to predict the biosynthetic gene clusters (BGCs) with the Cluster Finder Algorithm (Blin et al. 2019)

Antimicrobial gene (AMR) and virulence factors (VF) analysis

The CARD database was searched against the assembled genome of S-33 using a homology-based approach (BLASTX) to identify the presence of any AMR genes in the bacterial genome (Mcarthur et al. 2013). The BLAST output was filtered based on a minimum of 80% identity and subject protein coverage to retain high-confidence results. The VFDB database was searched against assembled genome sequence on criteria of a minimum 80% identity using a homology-based approach (BLASTX) to identify the presence of any virulence genes in strain S-33 (Kumari et al. 2023).

Comparative genome analysis

The circular genome comparison of the assembled genome of S-33 was performed against the publicly available reference genomes of *E. vonholyi*, *E. roggenkampii*, *E. quasiroggenkampii*, and *E. asburiae* using the BRIG (Blast Ring Image Generator) Tool (Alikhan et al. 2011). BLAST was performed on five characteristic *Enterobacter* genomes, which were constructed using NCBI local BLAST-2.10.1. The comparison and annotation of orthologous gene clusters were analyzed using Orthovenn2 in the genomes of *Enterobacter* sp. S-33, *E. mori* LMG 25706, *E. cancerogenus* ATCC 35316, *E. cloacae* ATCC 13047, and *E. cloacae* GS1 (Xu et al. 2019).

Data deposition

The genome sequence of S-33 is available at NCBI with the BioProject accession no. PRJNA860059.

Results

Isolation and characterization

Among the twenty isolated colonies, a colony that showed lactose fermentation on McConkey agar was further used for detailed characterization. The isolate was found to be Gram –ve and it showed the negative results for IMViC, amylase, pectinase, and positive results for lipase, cellulase, and catalase test (Supplementary Table 1). Among the tested various carbon sources, strain was able to utilize

lactose, xylose, maltose, galactose, melibiose, sorbitol, glycerol, D-arabinose, citrate, malonate, ONPG, and mannoside (Supplementary Table 1). The isolate showed higher sensitivity (20 to 25 mm) against tetracycline, ciprofloxacin, and moderate sensitivity (12 to 18 mm) to streptomycin, ampicillin, gentamycin, vancomycin, kanamycin, and fluconazole. The strain was found to be against voriconazole and erythromycin (Supplementary Table 2). The test isolate showed good antagonistic activity against *S. typhi*, *E. coli*, and moderate activity against *B. subtilis* and *S. aureus*. Against the tested fungal strains, strain showed good activity against *A. niger*, *M. gypseum*, *H. gypsium*, and moderate against *P. citrium* (Supplementary Table 3). The test isolate showed swimming, swarming, and twitching motility behavior (Fig. 1). The isolate showed the optimum growth in pH 6.0 to 9.0. Strain S-33 showed optimum growth up to 7% NaCl, while it could tolerate up to 10% NaCl. It showed the growth up to 45 °C of temperature. The 16S rRNA of isolate S-33 was submitted to NCBI Genbank database and accession no. OM943077 was assigned. The phylogenetic analysis of S-33 was performed by the neighbor-joining method between related-type strains of the *Enterobacter* species, which showed the phylogenetic position of strain S-33 (marked in bold) in the genus *Enterobacter*. In the phylogenetic tree, S-33 and different species of *Enterobacter* clustered in the same branch, which showed the closest genetic relationship. Based on phylogenetic analysis and 16S rRNA sequence similarity, it is obvious that strain S-33 may be a new species of *Enterobacter* (Supplementary S1).

T6SS genes

A 483 bp of the Hcp gene was amplified and sequenced (Fig. 2a, lane A). The phylogenetic analysis showed that Hcp of S-33 was closely matched to *E. cloacae* WP 094934690.1 and *E. asburiae* WP 143346857.1 (Fig. 2b). The partially amplified Hcp was submitted to GenBank with the accession number ON809441. Similarly, the VgrG gene of S-33 was

partially amplified (1.9 kb) (Fig. 2a, lane B), sequenced, and submitted to GenBank with accession number ON809442. The phylogenetic analysis of the VgrG gene showed its closest match to other *Enterobacter* spp (Fig. 2b).

Efflux and outer membrane protein (OMP) genes

The PCR reaction resulted in the partial amplification of 780 bp of *acrA* and 840 bp of the *acrB* gene (Supplementary S2), respectively. The product was sequenced and submitted to GenBank with the accession nos. ON854912 and ON854913, respectively. Similarly, partial segments of OMP genes such as *OmpA* (720 bp), *OmpC* (660 bp), and *OmpF* (660 bp) were amplified. The amplified genes were sequenced and submitted to GenBank with accession nos. ON854916 (*OmpA*), ON854915 (*OmpC*), and ON854917 (*OmpF*), respectively.

Hydrophobicity and co-aggregation assay

The cell hydrophobicity of S-33 was measured using the BATH (bacterial adherence to hydrocarbon) method. The adherence percentage of S-33 in hexane, hexadecane, toluene, and xylene was 28%, 21%, 37%, and 45%, respectively (Supplementary S3a). The co-aggregation percentage of S-33 with *B. subtilis* at 2 h, 4 h, 6 h, and 8 h was 4.5%, 6.2%, 9.2%, and 28%, respectively. Against *S. aureus*, the co-aggregation was 3.9%, 6.9%, 9.7%, and 31.5% at 2 h, 4 h, 6 h, and 8 h, respectively (Supplementary Fig. S4B). Similarly, with *E. coli*, S-33 showed the co-aggregation percentage of 7.1%, 9.1%, 12.5%, and 44.6% at 2 h, 4 h, 6 h, and 8 h, respectively, and with *S. typhi* was 8.2%, 9.5%, 11.8%, and 40.4%, respectively (Supplementary S3b).

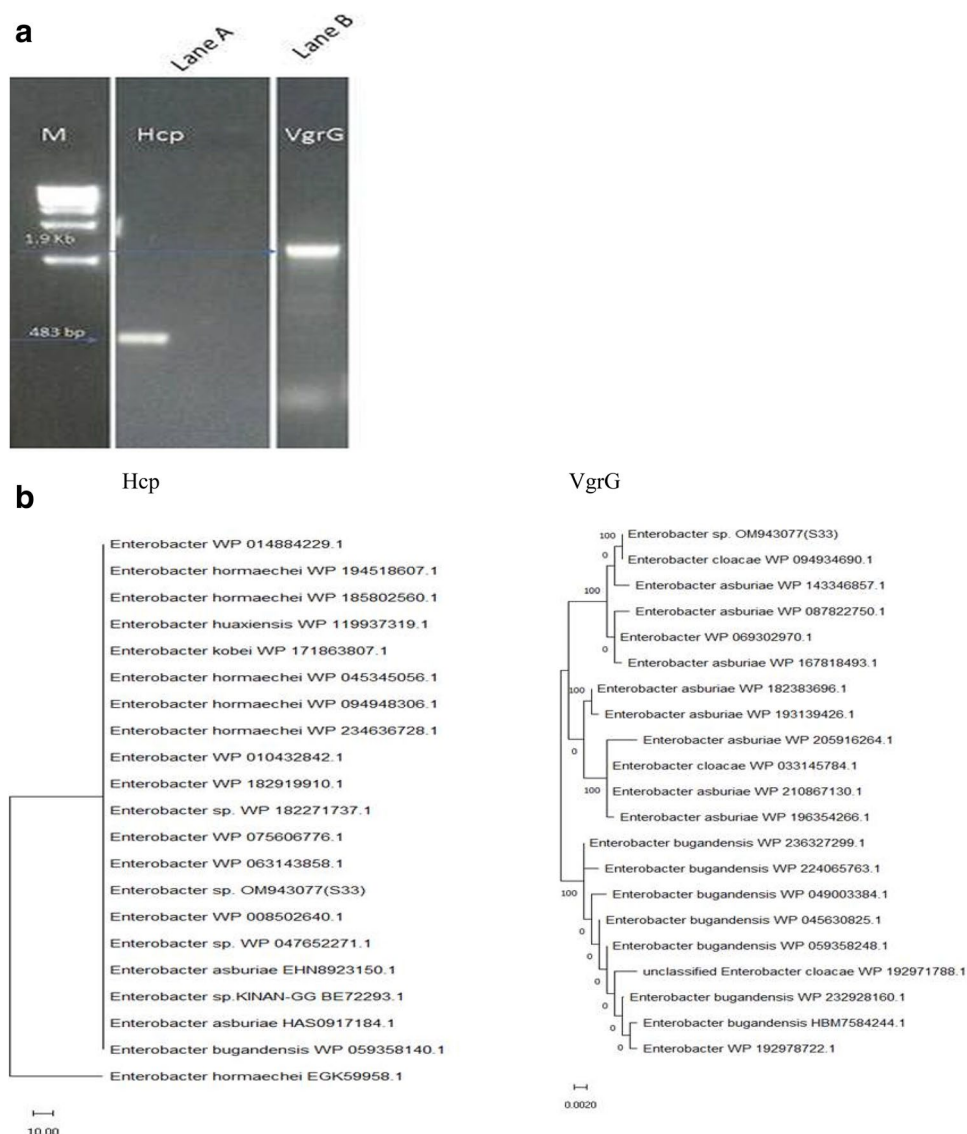
Genome analysis

A total of 8.5 million sequencing reads were used to create the assembled genome of S-33. The assembly was validated



Fig. 1 Test of motility shown by *Enterobacter* sp. S-33; **A** swimming, **B** swarming, and **C** twitching

Fig. 2 **a** PCR amplification of T6SS genes, e.g., Hcp (hemolysin coregulated protein) and VgrG (valine-glycine repeat protein G) (images is from the different gel). The PCR product was run on 1% agarose gel electrophoresis in 1× TAE buffer and visualized in a UV-gel documentation system; **b** phylogenetic analysis of T6SS effector proteins, Hcp and VgrG



using the NCBI-NR Blast program, where it showed the maximum homology with other *Enterobacter* sp. strains. The G + C content (55.63%) of S-33 was very similar to that of *E. vonholyi*. A total of 83 tRNA and 25 rRNA were annotated in the genome (Table 2). Hybrid assembly was performed using Unicycler v-0.48 tool for assembly of Illumina short and long reads which resulted in a single contig assembly of 4.6 Mb. A single circular chromosome of 4.6 Mbp in size with no plasmids was obtained (Fig. 3). Further, gene and protein prediction from the draft genome using the Prokka v1.14 tool identified a total of 4219 proteins. With the protein sequences, Orthovenn2 exhibited the ortholog clusters shared by all five species were 2592, 1001 shared by four species, 397 and 581 shared by three and two species, respectively. A total of 260 ortholog gene clusters were specific to only a single genome. Out of the 260 gene clusters, 55 belonged to *Enterobacter* sp. S-33, 21 to *E. cancerogenus*

Table 2 Genome feature of *Enterobacter* sp. S-33

	Annotated genome features
Contigs	1
GC content	55.63%
CDS	4,343
tRNA	83
rRNA	25
Proteins with functional assignment	4219
Hypothetical proteins	486
Proteins with GO assignment	1010
Proteins with pathway assignment	873

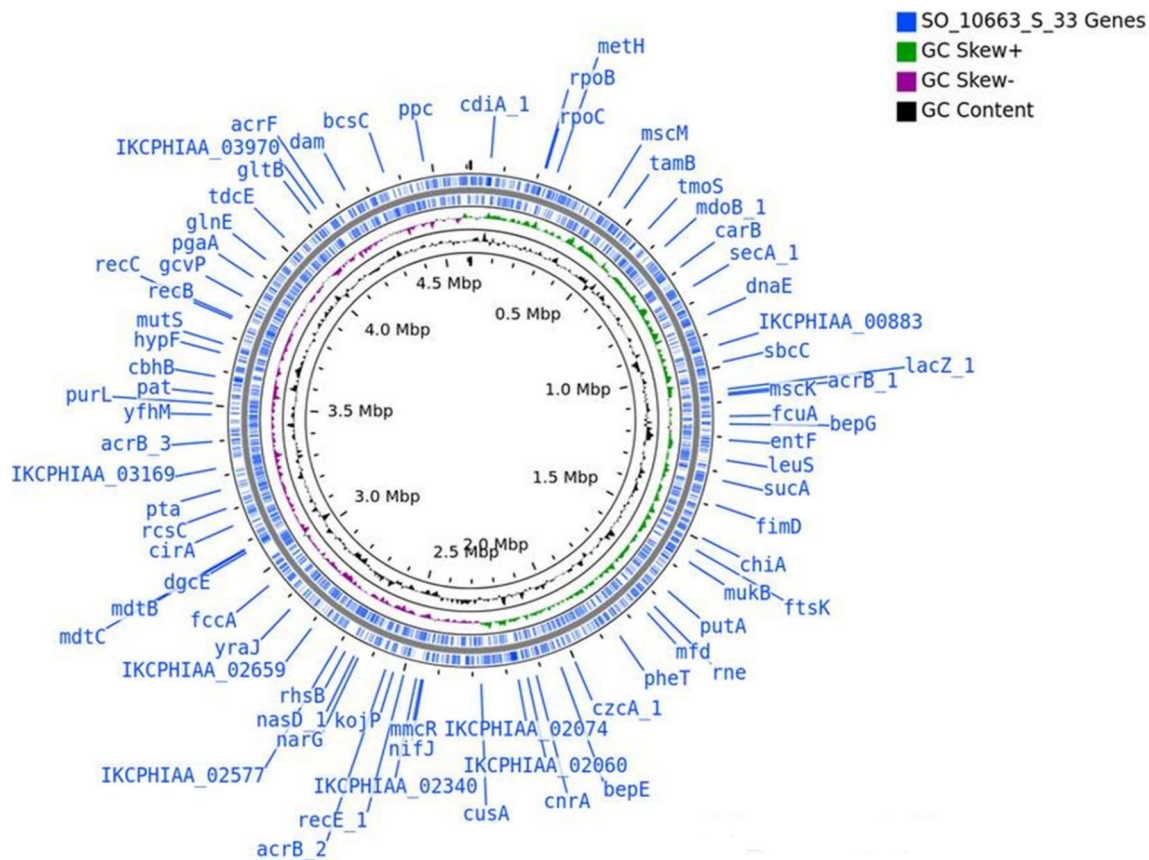


Fig. 3 Circular genome map of *Enterobacter* sp. S-33 (marked as SO_10663_S_33). The distribution of the circle from the outside to the inside indicates the genome size forward CDS, reverse CDS, repeat sequences, GC-skew⁺, GC-skew⁻, and GC-content

ATCC 35316, and 135 to *E. cloacae* ATCC 13047, whereas 8 and 41 specific gene clusters were from *E. cloacae* GS1 and *E. mori* LMG 25706, respectively (Supplementary S4).

Average nucleotide identity

ANIb was calculated using reference genomes of *Enterobacter* (Supplementary Table 4) to confirm the identity of the S-33 genome. ANIb analysis showed the highest similarity of S-33 to *Enterobacter quasiroggenkampii* WCHECI-1060^T (94.10%), *Enterobacter quasihormaechei* WCHEs120003^T (92.20%), and *Enterobacter cloacae* subsp. *cloacae* ATCC 13047^T (91.44%). (Supplementary Table 4). The values were higher than the species cut-off threshold value of 95%, and therefore identified as *Enterobacter* sp. The ANI values with other *Enterobacter* species were 90.16 to 77.41%. The analyses of ANI and isDDH were all lower than the species-delineating threshold (95% and 70%), indicating that strain S-33 does not belong to an existing species and would be a unique new species in the *Enterobacter* genus. As shown in Supplementary S5, the whole genome-based phylogenomic tree of S-33 and related-type strains of the *Enterobacter* genus showed the monoclade position of strain

S-33 with *Enterobacter quasiroggenkampii* and *Enterobacter quasihormaechei*, suggesting the closest genetic relationship between them, which was similar to the results of 16S rRNA phylogeny. Combined with the results of comparative genomic analysis, S-33 would be a novel candidate species in the *Enterobacter* genus.

RAST functional annotation

Enterobacter sp. S-33 genome was uploaded for annotation to RAST, and subsystem coverage and non-subsystem coverage generated are 62% and 38%, respectively (Fig. 4A). RAST analysis predicted the subsystem category distribution (Fig. 4B) and subsystem feature counts (Fig. 4C). The top three subsystem features of S-33 are carbohydrates (562 genes), amino acid derivatives (488 genes), and protein metabolism (294 genes) (Fig. 4). The other subsystem categories related to vitamins and cofactors (268 genes), RNA metabolism (213 genes), cell wall and capsules (207 genes), membrane transport (199 genes), and virulence genes (112 genes) were observed (Fig. 4).

RAST-based functional annotation found several key elements associated with the adaption of *Enterobacter* sp.

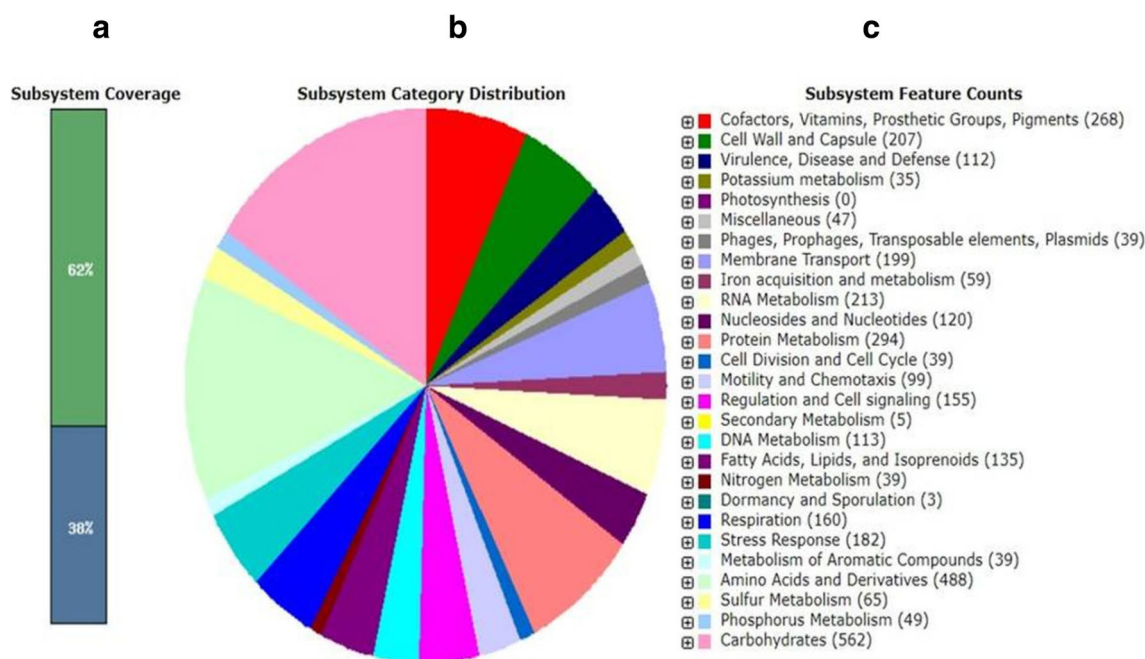


Fig. 4 Pie chart representing RAST subsystem category in the *Enterobacter* sp. S-33 genome. The most abundant systems on the category level are shown in the left pie chart, whereas the right column shows the counts of features

S-33 to various niches. Strain S-33 found several elements associated with the osmoregulatory systems including the osmotic stress cluster (4 genes), osmoregulation (5 genes), osmoprotectant ABC transporter (4 genes), osmoregulatory periplasmic glucans (6 genes), and choline betaine biosynthesis clusters (8 genes). Among the osmotic stress cluster, genes belonging to osmotic inducible lipoprotein (OsmB), heat shock protein (YciM), phosphatidylglycerophosphatase B (PgpB), and translationally regulatory protein (YciT) were identified. Among the osmoprotectants, ABC transporter, ATP binding subunit (YehX), osmoprotectant binding protein (YehZ), osmoprotectant inner membrane protein (YehW), and permease protein (YehY) were identified (Fig. 4).

Gene ontology

To understand the putative biological functions of the classified proteins, gene ontology was investigated (Fig. 5). Among the biological processes, the largest number of proteins was observed for membrane components (13.18%), followed by methylation and translation (2.67%), transcriptional regulation (1.67%), cell division, transmembrane transport (1.24%), etc. Among the cellular component, a higher number of proteins were reported for cytoplasmic proteins (8.93%), followed by plasma membrane (7.93%), and ribosomal proteins (2.15%). The ABC-transporter and cell-outer membrane proteins were 0.96% and 0.86%, respectively. In molecular function, higher

numbers of proteins belonged to ATP-binding (8.98%), transferase activity (7.16%), hydrolase activity (6.92%), DNA-binding (5.97%), metal-binding (4.2%), etc. KEGG analysis identified the proteins belonging to the various metabolic pathways (Fig. 6). The highest number was recorded for signaling and cellular processes proteins, followed by genetic information processing and carbohydrate metabolism.

The various BGCs in the strain S-33 genome were annotated using the antiSMASH database. In terms of biosynthetic paradigms, BGCs comprised non-ribosomal polypeptides (NRPS) (regions 1,185,553 to 1,229,410), O-antigen containing thiopeptide (regions 1,531,486 to 1,557,797), siderophore-like aerobactin (regions 2,971,238 to 2,985,666), and aryl polyenes (regions 4,287,759 to 4,331,352) (Supplementary S6 A). The NRPS cluster of S-33 showed highest 60% similarity to bacillibactin of *Bacillus velezensis* FZB42, 42% to amonabactin of *Aeromonas hydrophila* subsp. *hydrophila* ATCC 7966, followed by 30% similarity with turnerbactin of *Teredinibacter turnerae* T7901, and 25% with amychelin of *Streptomyces* sp. AA4 (Supplementary S6B). A gene cluster homologous to the BGCs of aerobactin was found in the S-33 genome, which showed 44% similarity with *Pantoea ananatis* and *Xenorhabdus szentirmaii* DSM 16338 genome (Supplementary S6C). Similarly, the arylpolyene cluster of the S-33 genome showed the highest similarity of 100% to *X. doucetiae* and 94% to *E. coli* CFT073 genome (Supplementary S6D).

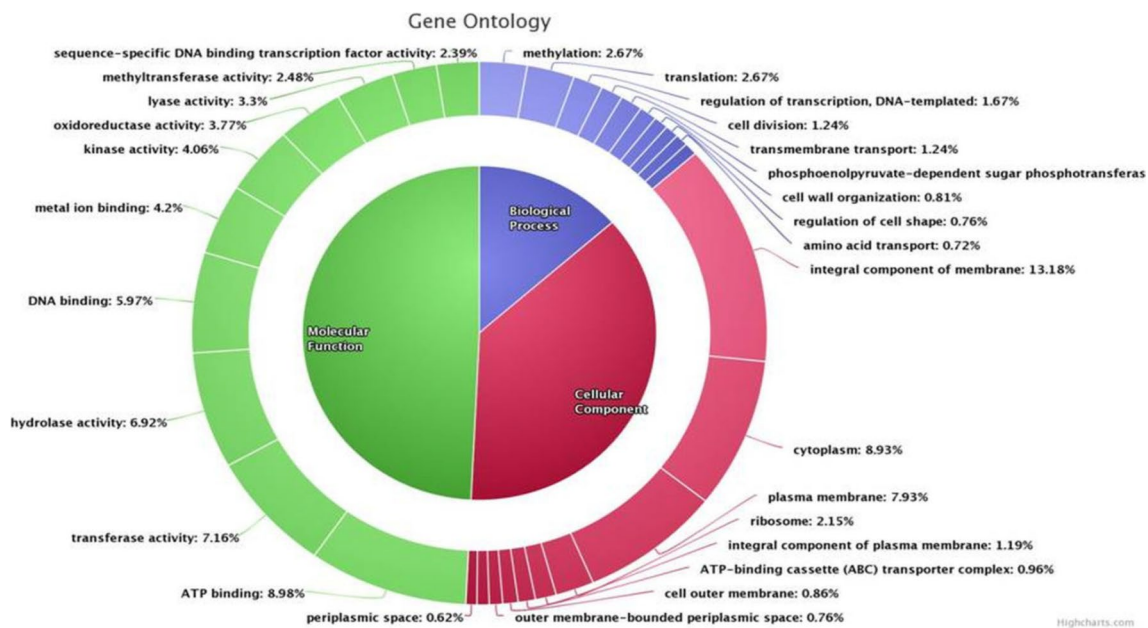
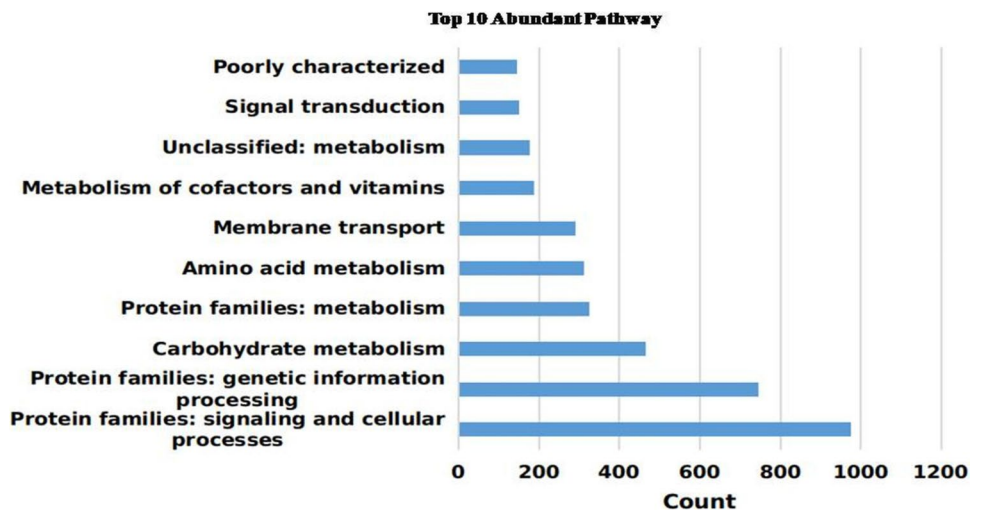


Fig. 5 Gene Ontology (GO) analysis in the *Enterobacter* sp. S-33 genome. The analysis was performed by using the gene sequence of S-33 in Blast2Go program and functions were assigned into 3 different categories including molecular, cellular, and biological function

Fig. 6 Pathway analysis was performed using Kyoto Encyclopedia of Genes and Genomes (KEGG) Automatic Annotation Server (KAAS) database. KAAS database is used for functional annotation of genes by BLAST comparisons against KEGG-GENES database



Secretory pathway genes and AMR analysis

Genomic analysis identified the T1SS-associated SymE protein, the T2SS GspC protein, and T4SS Rhs protein. In the genome analysis, we identified the hallmark of T6SS proteins such as Hcp, VgrG, and other T6SS-associated components such as TssA, TssH, TssL, TssM, etc. Genome analysis identified a single cluster of T6SS in the S-33 genome. Based on 70% identity and subject protein coverage to retain high confidence results, a total of 54 AMR proteins were found. The different classes of AMR and gene numbers have been summarized in Supplementary S7. Gene families like rifamycin-resistant beta-subunit of RNA polymerase

(rpoB), resistance-nodulation-cell division (RND) antibiotic efflux pump, triclosan-resistant gyrA, antibiotic-resistant cyaadenylate cyclase, fluoroquinolone-resistant gyrB, fluoroquinolone-resistant parC, ATP-binding cassette (ABC) antibiotic efflux pump, fluoroquinolone-resistant parE, antibiotic-resistant ptsI phosphotransferase, pmr phosphoethanolamine transferase, etc., were found in the blast output results (Fig. 7a). These belong to resistance mechanisms such as antibiotic target alteration, reduced permeability to the antibiotic, antibiotic target replacement, and antibiotic efflux.

A 500 bp upstream and downstream coordinates for the given 80 AMR genes were fetched from the draft assembly

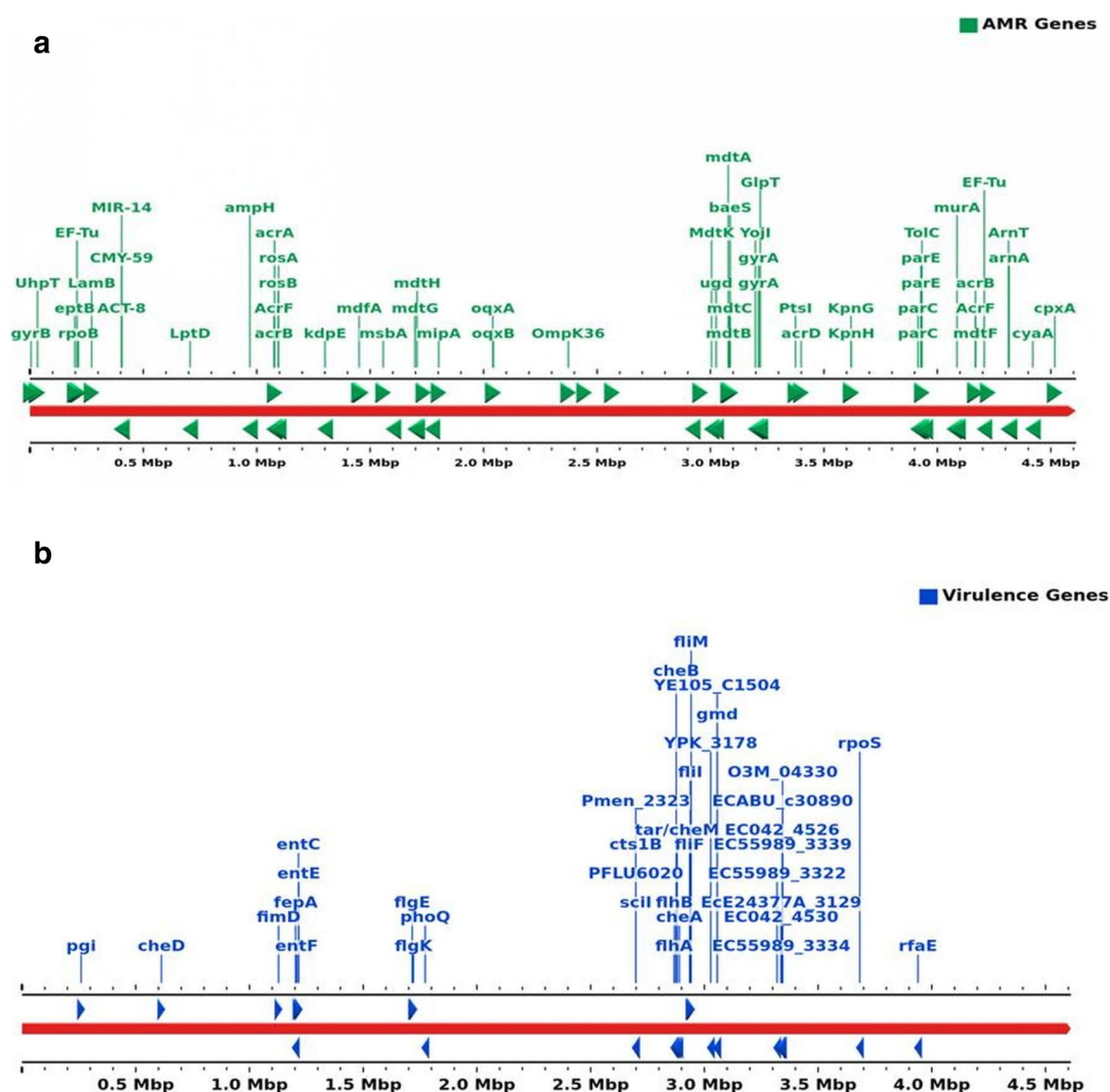


Fig. 7 **a** Identification of antimicrobial resistance genes (AMR) in *Enterobacter* sp. S-33 using CARD database through BLASTX program with minimum 80% gene identity; **b** identification of virulence

factor genes (VFs) in *Enterobacter* sp. S-33 using VFDB database through BLASTX program with minimum 80% gene identity

genome for identification of mobile genetic elements using a homology-based approach (BLASTX) against MGE protein sequences (ACLAME database). The blast output was filtered based on criteria of minimum 70% identity and subject protein coverage to retain a high confidence result, where we found 1 mobile genetic element in the 500 bp upstream of AMR gene *nfsA*. The mobile element was identified as glutaredoxin 1. We have also found mobile gene elements such as outer membrane lipoprotein BLC protein and a hypothetical protein in the 500 bp downstream of 4 AMR genes (*murA* transferase, MIR beta-lactamase, ACT beta-lactamase, and CMY beta-lactamase).

VF analysis

The blast output was filtered based on criteria of minimum 70% identity and subject protein coverage to retain high confidence result, where a total of 49 virulence proteins were found to have different functions such as enterobactin synthase multienzyme complex component, putative type VI secretion protein, outer membrane usher protein SfmD, ferrienterobactin, chemotaxis protein, 2,3-dihydroxybenzoate-AMP ligase, flagellum-specific ATP synthase, phosphomannomutase, isochorismate synthase 1, etc. (Fig. 7b). A 500 bp upstream and downstream coordinates for given 80 VF genes

were fetched from assembled genome for identification of mobile genetic elements using homology-based approach (BLASTX) against MGE protein sequences (ACLAME database). The blast output was filtered based on criteria of minimum 70% identity and subject protein coverage to retain high confidence result, where we found 4 mobile elements in 500 upstream of the VF-associated genes EC55989_3339, ECABU_c30890, phoQ sensor protein, and O3M_04330. We have also found mobile gene elements such as chemotaxis and hypothetical proteins in the 500 downstream of 6 VF-associated genes EC042_4530, EC55989_3334, EcE24377A_3129, cheA, and cheB (belonging to secretion, chaperone, and chemotaxis protein family).

Comparative genome analysis

The circular genome comparison of S-33 was carried out against 4 reference genomes, i.e., *E. vonholyi* strain E13, *E. roggenkampii* strain DSM 16690, *E. quasiroggenkampii* strain WCHECL1060, and *E. asburiae* strain ATCC 35953 using the BRIG (v 0.95) tool which helps to determine the homologous regions between sets of genomes. The output image shows similarity between the assembled genome and other reference sequences as a set of concentric rings, where BLAST matches are colored on a sliding scale indicating a defined percentage identity. A whole genome alignment-based identity and coverage between the assembled genome of the bacterial sample shows highest homology with the reference genome of *E. vonholyi* (98.41% identity and 93% coverage) followed by *E. roggenkampii* (95.82% identity and 90% coverage), *E. quasiroggenkampii* (94.84% identity and 91% coverage), and *E. asburiae* (95.35% identity and 87% coverage) (Supplementary S8). Around 87–93% of the regions from the assembled genome of the sample can be considered to be conserved with respect to the 4 reference genomes. Similarly, 7 to 13% of the regions can be considered variable regions with respect to the given reference genomes. Additionally, we compared the presence of virulence factors in these selected strains which showed that S-33 was having a higher number of genes involved in the secretion system, iron uptake, fimbrial adherence, toxin production, antiphagocytosis, and immune invasion (Supplementary S9).

Discussion

In recent years, *Enterobacter* spp. quickly developed resistance to various antibiotics and antibacterial drugs by gaining exogenous resistance genes. The development of antibiotic resistance by HGT (horizontal gene transfer) among *Enterobacter* sp. is of great concern worldwide (Davín-Regli and Pagès 2015). It increases the risk of antibiotic treatment

failure both in humans and animals, and also the chances of contamination and human public health consequences if transmitted by human pets (Lloyd 2007). Therefore, understanding the distribution of antimicrobial resistance among *Enterobacter* spp. is important from both veterinary and public health perspectives. However, all these processes require detailed characterization of *Enterobacter* spp. both at the biochemical and genomic levels. Therefore, in the present study, our aim was to investigate the detailed characterization of an environmental *Enterobacter* isolate to confirm all the antibiotic, virulence, and other related properties.

The emergence of drug resistance in *Enterobacter* spp. poses a challenge for effective treatment and therefore, an urgent need to develop effective vaccines is required. Safety assessment with respect to virulence factors, antibiotic resistance, and animal experimentation is an important factor in the choice of *Enterobacter* species (Paauw et al. 2008). The presence of antibiotic resistance genes in *Enterobacter* organisms is often organized on the plasmid or transposon-related, thereby, presents the potent risk of horizontal gene transfer (Rozwandowicz et al. 2018). The environmental isolate S-33 showed susceptibility to some clinically relevant antibiotics like tetracycline and ciprofloxacin; however, the strain was found to be resistant to voriconazole and erythromycin. The WGS (whole genome sequencing) analysis identified the various genes showing resistance to aminoglycosides, sulfonamides, and peptide antibiotics. In the WGS data, we found various ARGs like β -lactamase which are mainly responsible for the resistance mechanism of bacteria against β -lactam antibiotics.

To resist the antibiotics, bacteria utilize different mechanisms such as activation of membrane-bound efflux pumps, modification of drug site action, and destruction of antimicrobials (Blair et al. 2015). Drug efflux pumps confer an important role in the pumping of antibiotics from the cell and therefore are responsible for the multidrug resistance of Gram –ve bacteria (Nishino et al. 2009). The drug resistance in *Enterobacter* spp. was mainly attributed to the *acrAB* and *acrD* efflux pumps belonging to the RND super-family (Sun et al. 2014). In the test isolate S-33, multiplex PCR assay confirms the presence of efflux genes *acrA* and *acrB*. The search for OMPs revealed the presence of *OmpA*, *OmpC*, and *OmpX* in the test isolate under study, coinciding with a previous report suggesting these OMPs as integral part of Gram –ve bacterial membrane (Dupont et al. 2004). The outer membrane *OmpF*, *OmpC*, *Omp36*, *OmpK36*, and *OmpK37* genes are mainly responsible for porin-mediated antibiotic resistance by reducing permeability to β -lactams (Dupont et al. 2004; Viveiros et al. 2007). The overproduction of the *OmpX* outer membrane protein decreased the bacterial susceptibility to a wide range of antibiotics (Mishra et al. 2020). Moreover, the presence of these OMPs promote resistance against multiple classes of antimicrobial agents,

such as β -lactams, fluoroquinolones, tetracyclines, and rifamycin, suggesting that their presence provides broader resistance to multiple antibiotics (Song et al. 2014). A study demonstrated the role of *E. cloacae* OmpX in the invasion of epithelial cells, which suggests its possible role in virulence (de Kort et al. 1994).

The prevalence of virulence factors in S-33 reinforced the invasive nature and pathogenic potential and also highlighted the importance of additional studies to investigate the in vitro and in vivo roles of these traits, in order to provide more clear details about the virulence and infection potential of isolated strain. Previous studies showed that some *Enterobacter* species, namely, *Enterobacter aerogenes*, *E. cloacae*, and *E. hormaechei* with increased mortality pose a serious threat to human society (Akbari et al. 2016; Anastay et al. 2013). An environmental isolate, *E. cloacae* SBP-8, possessed several virulence factors and its pathogenicity was established by using *Caenorhabditis elegans* as a model host (Khan et al. 2020). The soil isolate SBP-8 also showed the presence of T6SS (Singh et al. 2017) and its infection significantly affected the *C. elegans* survival, physiology, and reproduction (Khan et al. 2020).

Bacterial surface hydrophobicity is often associated with adhesiveness and varies from organism to organism, and also from strain to strain. A high level of cell hydrophobicity was observed for the S-33 with the tested hydrocarbons. Higher hydrophobicity was recorded with the xylene which is in agreement with the previous report of Dillon et al. (1986). The difference in adherence ability to different hydrocarbons has been observed for some organisms (Xu et al. 2018). Also, the cell hydrophobicity of different pathogenic microorganisms varies with the growth stage. A previous study reported the higher surface hydrophobicity of *P. aeruginosa* in the early to mid-log exponential phase as compared to the stationary phase (Jana et al. 1999).

The property of aggregation often associated with adherence is considered a prerequisite for bacterial colonization and is often associated with specific interactions between the bacterial cell surface and the host surface (Collado et al. 2008). The respective genes encoding S-layer protein (Slp), adhesion, and fibronectin-binding protein that assist in binding to the extracellular matrix were mapped in the S-33 genome. Similarly, flagellum, the thin filament extension from the bacterial surfaces, acts as adhesion to facilitate adherence to the hydrophobic surface of the intestine such as mucin (Haiko and Wikström 2013). The WGS analysis identified the various genes belonging to flagellin (FliC/FliJ), flagellar hook-associated (FlgL/FlgE), flagellar hook-length controller (FliK), flagellar basal-body rod (FlgG), flagellar biosynthesis (FlgF), etc., which suggest the possible pathogenicity of the S-33 strain.

In their natural environment, there is a possibility of limited nutrient availability; therefore, bacteria evolve

their metabolic system to support the utilization of different substrates. Interestingly, ten inositol catabolism genes were recorded in addition to pentose phosphate, lactose, and sucrose utilization genes. Similarly, a maltose, lactate, and maltodextrin utilization gene was also annotated in the S-33 genome, supporting its ability to utilize various carbon sources. In all 562 annotated genes related to carbon metabolism, we observed central 114 genes for central carbohydrate metabolism, 9 genes for amino sugars, 29 genes for organic acids metabolism, 10 N-acetylglucosamine utilization genes, 5 fructoselysine utilization pathway genes, 5 dehydrogenase genes, and 14 for sugar alcohols. Similarly, forty-nine genes related to phosphorus metabolism and sixty-five sulfur metabolism genes were identified in S-33 genome. Several iron transport systems including ferric hydroxamate ABC transporter PhuCDBA, 14 ferric enterobactin transporter proteins (EntA, EntE, EntD, EntB, Fes, EntS, EntF, YbdZ, FepC, FepD, FepG, FepE, FepB, EntC), ferric reductase protein PhuF, and a periplasmic binding protein TonB were annotated in the S-33 genome.

To survive in different environments, bacteria develop several strategies including genetic adaptation, mutations, and gene transfers leading to the development of new traits that finally favored the bacteria to survive within the new ecological niche (Maurelli 2007). Genome mining suggested that *Enterobacter* sp. S-33 contained a repertoire of genes favoring osmolyte tolerance that could function to aid survival. One hundred and eighty-two annotated genes were identified as being involved in various stress responses. Their presence in the genome provides insight into the adaptation of S-33 to survive under varying environmental conditions. Interestingly, the osmoprotectant ABC transporter, glycine betaine MFS transporter, betaine/carnitine/choline transporter consisting of high-affinity choline uptake gene, betaine aldehyde dehydrogenase, and a choline dehydrogenase gene was annotated in S-33 genome. Several other genes linked to osmotic stress including five osmoregulation genes, five osmotic stress cluster genes, and seven osmoregulated periplasmic glucan genes were identified. The biosynthetic potential was demonstrated by the genome mining within the S-33 genome which showed the presence of NRPS clusters that produce molecules with antibiotic activity and with competition between different microbes (Gulick 2017; Steinke et al. 2021). The gene clusters with aerobactin biosynthetic pathways were observed that is involved in the biosynthesis of the siderophores (Bailey et al. 2018).

Several pathogenesis-associated genes such as T1SS, T2SS, T3SS, T4SS, and T6SS- encoding secretion system proteins responsible for disease in both animals and plants were identified (Green and Mecsas 2016). Genome annotation identified the T1SS family toxin SymE, T2SS protein GspC, protein E, T2SS core protein G, and other T2SS-associated proteins including protein J/K/L/M. Gram –ve

bacteria utilize T1SS to secrete the proteins via translocator protein complexes using (i) cytoplasmic membrane proteins/ or membrane fusion proteins, (ii) ATP-binding cassette, and (iii) outer membrane proteins. Following activation, ABC exporter and T1SS proteins provide substrate specificity for activation, leading to the secretion of proteases and C-terminal secretion signals (Delepelaire 2004). T3SS protein included the invasion protein *lagB*, whereas T4SS included the *Rhs* protein, *DotU* family protein, etc. The T3SS is present in many Gram –ve bacterial pathogens, plants, and animal symbionts, and comprises the cytoplasmic, integral, and outer membrane proteins (Abby and Rocha 2012; Magdalena et al. 2002). The T4SS resembles the conjugation system of bacteria and based on structural components, categorized into type 4A (T4ASS) or type 4B (T4BSS) (Vot et al. 2012). This machinery is directly involved in the transfer of effector proteins to the host cytosol through a central pore.

The T6SS transfers effector proteins directly into host cells (i.e., eukaryotic and prokaryotic) in a contact-dependent manner. The T6SS imparts various functions including virulence, host-pathogen, and microbe-microbe interactions. The major targets of T6SS effectors include the cell wall, membrane, and nucleic acids (Records 2011). In the test isolate, among T6SS proteins, we identified the Hcp family effector, tip protein *VgrG*, TssH encoded-ATPase, baseplate subunit protein TssF/TssE, T6SS-associate protein *ImpH/TagH*, amidase immunity proteins, contractile sheath large subunit *EvpB* family protein, membrane subunit TssM, and T6SS proteins TssA/TssL/*ImpJ*. Recently, two T6SS clusters were identified in *E. cloacae*, i.e., T6SS-1 and T6SS-2, which are involved in bacterial competition and virulence activity (Chen et al. 2019; Journet and Cascales 2016). The *E. cloacae* T6SS-1 and T6SS-2 are involved in the inter-bacterial competition, biofilm formation, adherence to cells, and intestinal colonization (Soria-Bustos et al. 2020). Similarly, T6SS was revealed in the genome of *E. mori* by whole-genome analysis which was related to bacterial virulence (Green and Mecsas 2016). In summary, the newly isolated strain S-33 showed several pathogenesis-associated genes and the presence of multiple AMRs. Therefore, future studies must be welcomed to test the pathogenicity of this isolate by using a suitable model system which will increase knowledge about the safety aspects of environmental isolates belonging to the *Enterobacteriaceae* family.

Conclusion

In summary, the environmental isolate S-33 showed the presence of antibiotic and virulence genes. As repositories of AMRs, these bacterial groups have a great risk that needs more attention. To further improve our clear understanding, more extensive and multilocus studies are required to

analyze the phylogenetic relationships of different sources of *Enterobacter* and its closest relatives, which will shed more light on the transmission mechanism of AMR genes and plasmids. The findings of the current study are significant as they highlight the potential human health risk of environmental *Enterobacter* isolate. The test isolate showed a higher number of virulence genes as compared to other tested genomes, and hence future studies will be directed to elucidate the role of identified T6SS effectors, toxin, and other virulence genes for their any role in pathogenicity by using suitable model systems. In conclusion, genome analysis of S-33 highlighted a number of potential features that might be considered candidates for future studies to explore the understanding of the virulence mechanism deployed by this bacterium.

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Author contributions K.K. analyzed the data. Y.A. helped in the interpretation of data and editing of manuscript. R.P.S. designed the work and wrote the manuscript. All authors reviewed the manuscript.

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Data availability The datasets generated during and/or analyzed during the current study are available in the manuscript and from the corresponding author on reasonable request.

Declarations

Ethics approval This article does not contain any studies with human or animals performed by any of the authors.

Competing interests The authors declare no competing interests.

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